

Nexivarin 10 mg Tablets — eCTD Submission
Module 2 — CTD Summaries

Submission Type	NDA — 505(b)(1)	Sequence Number	0000
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PharmaCorp Inc.
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Section 2.5
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2.5 Clinical Overview

2.5.1 Product Development Rationale

Hypertension affects approximately 1.28 billion adults worldwide and is a major risk factor for cardiovascular and renal morbidity. Nexivarin hydrochloride offers a distinct pharmacokinetic advantage: a 30–50 hour elimination half-life enabling once-daily dosing. The full clinical development rationale, including disease background and unmet need, is provided in **Section 2.5.1** and the Clinical Overview Appendices.

The epidemiology of hypertension and therapeutic landscape are referenced in Module 5, Section 5.4 (Literature References); see References [1] and [3] in Section 5.4 for the 2017 ACC/AHA and 2018 ESC/ESH hypertension guidelines.

2.5.2 Overview of Biopharmaceutics

An absolute bioavailability study (NXV-BA-01) confirmed oral bioavailability of 78.2% (CV 12%). Food had no clinically significant effect on AUC or C_{max}. Nexivarin tablets may be administered without regard to meals. The complete biopharmaceutics study report is in **Section 5.3.1** (Module 5); the biopharmaceutics summary is in **Section 2.7.1**.

Refer to Section 5.3.1 (Module 5) for the complete bioavailability and food-effect study report (Study NXV-BA-01). The food-effect statistical results (AUC ratio 1.04; 90% CI 0.96–1.12) are listed in Table 1 of Section 5.3.1. As shown in Figure 5.3.1-1, mean plasma concentration-time profiles under fed and fasted conditions are superimposable.

2.5.3 Overview of Clinical Pharmacology

Steady-state PK parameters following 7-day once-daily dosing of nexivarin 10 mg are summarized in Table 2.5-1 below. The full clinical pharmacology summary and population PK analyses are provided in **Section 2.7.2**. Individual clinical pharmacology study reports are in Module 5 (Section 5.3.1).

PK Parameter	Value	Notes
C _{max} (steady-state)	18.2 ± 3.4 ng/mL	After 7 days
AUC _{0–24} (steady-state)	412 ± 68 ng·h/mL	
T _{max}	6–10 h	Median 8 h
t _{1/2} (terminal)	38.4 ± 6.2 h	Enables once-daily dosing
V _d /F	312 ± 48 L	Extensive tissue distribution
Cl/F	24.2 ± 4.1 mL/min	Hepatic elimination
Protein binding	97.5%	Primarily albumin
Accumulation ratio	2.1	At steady-state

Data listed in Table 2.5-1 are derived from the integrated population PK analysis (see Section 2.7.2 for the complete population PK model report and Figure 2.7.2-1 for the goodness-of-fit plots). Individual study-level PK data are provided in Section 5.3.1 (Module 5). The clinical AUC_{0–24} of 412 ng·h/mL is the reference value used for nonclinical safety margin calculations (refer to Section 4.2.3.2, Module 4).

2.5.4 Overview of Efficacy

Efficacy was established in two pivotal Phase III studies. Full study reports are in Module 5 (Sections 5.3.3.2 and 5.3.3.3); the integrated efficacy summary is in **Section 2.7.3**.

Study NXV-301 (n=892): 12-week, double-blind, placebo-controlled. Primary endpoint: mean change from baseline in SeDBP at Week 12. Nexivarin 10 mg significantly reduced SeDBP vs. placebo (–12.8 ± 0.6 vs. –4.2 ± 0.6 mmHg; LS mean difference –8.6 mmHg; 95% CI: –10.1 to –7.1 mmHg; p<0.0001). As shown in Table 2.5-2 below, all secondary endpoints

were also met. The full clinical study report is provided in **Section 5.3.4.2** (Module 5).

Endpoint	Nexivarin 10 mg	Placebo	LS Mean Diff (95% CI)	p-value
SeDBP change (mmHg)	-12.8 ± 0.6	-4.2 ± 0.6	-8.6 (-10.1 to -7.1)	<0.0001
SeSBP change (mmHg)	-18.4 ± 1.2	-7.3 ± 1.1	-11.1 (-13.8 to -8.4)	<0.0001
Responder rate (SeDBP <90 mmHg)	68.4%	31.7%	OR 4.38 (3.10–6.18)	<0.0001
Normalization (SeDBP <85 mmHg)	49.6%	18.4%	OR 4.26 (2.92–6.22)	<0.0001

As shown in Table 2.5-2, nexivarin 10 mg met all primary and key secondary efficacy endpoints in Study NXV-301. Full efficacy data, including time course of blood pressure reduction (Figure 5.3.4.2-1) and forest plots for subgroup analyses, are provided in Section 5.3.4.2 (Module 5). Study NXV-302 non-inferiority data are provided in Section 5.3.3.3 (Module 5); as described in that section, nexivarin 10 mg met the NI margin vs. amlodipine 10 mg (LS mean difference -0.4 mmHg; 95% CI: -1.6 to 0.8 mmHg).

2.5.5 Overview of Safety

A total of 1,874 patients received nexivarin 10 mg across all controlled studies. The integrated safety summary is presented in Table 2.5-3. The complete integrated safety data are provided in **Section 5.3.4** (Module 5 — Integrated Summary of Safety).

Parameter	Nexivarin 10 mg (N=1874)	Placebo (N=923)	Amlodipine 10 mg (N=450)
Any TEAE	62.4%	52.8%	65.1%
Serious AEs	2.8%	2.1%	3.1%
Discontinuation due to AEs	3.4%	1.6%	4.2%
Peripheral Edema	14.6%	1.8%	22.3%
Headache	7.3%	7.8%	7.9%
Dizziness	3.4%	1.5%	3.8%
Flushing	2.6%	0.5%	4.1%
Deaths	0%	0.1%	0%

As depicted in Table 2.5-3, the incidence of peripheral edema with nexivarin 10 mg (14.6%) is notably lower than that observed with amlodipine 10 mg (22.3%) in the head-to-head controlled study (NXV-302), representing a clinically meaningful tolerability advantage. The complete adverse events data by system organ class (SOC) and preferred term (PT) are provided in Section 5.3.4.3 (Module 5). Adverse events of special interest are listed in Table 1 of Section 5.3.4.3. Deaths are described in Section 5.3.4.4 (Module 5); no deaths were related to nexivarin.

2.5.6 Benefit/Risk Assessment

The benefit/risk profile of nexivarin 10 mg for the treatment of hypertension and stable angina pectoris is favorable. The drug provides robust antihypertensive efficacy (as established in Module 5, Sections 5.3.3.2 and 5.3.3.3), a lower incidence of peripheral edema than amlodipine (see Section 5.3.4), and a predictable PK profile (see Sections 5.3.1 and 2.7.2) supporting reliable once-daily dosing. The nonclinical safety data (Module 4) support adequate safety margins at the proposed clinical dose. No new safety concerns have been identified that would preclude approval.

The integrated benefit/risk discussion references: nonclinical safety margins (provided in Section [4.2.3.2](#), Module 4); clinical efficacy data (provided in Sections 5.3.3.2 and [5.3.3.3](#), Module 5); and integrated safety data (provided in Section 5.3.4, Module 5). The labeling proposed in Section [1.3.1](#) (Module 1) reflects this benefit/risk assessment.